

SEEING WITHOUT LIGHT

The resolution of a microscope is limited by wavelength. When it comes to observing tiny objects, matter waves (which have shorter wavelengths) can be more helpful than light. The wavelength of an electron is up to 100,000 times shorter than that of visible light, allowing for the imaging of up to 100,000 times smaller objects, such as viruses. Scanning electron microscopes use beams of accelerated electrons to illuminate objects, bent by magnetic fields rather than glass lenses.

SECONDARY
DETECTOR

A secondary electron detector attracts and registers electrons emitted from the sample.

Beaming electrons

A focused electron beam scatters from atoms on an object's surface, producing signals that allow for a very detailed image to be built.

REFLECTED
BEAMS

Some high-energy electrons are reflected or backscattered from atoms within the specimen.

